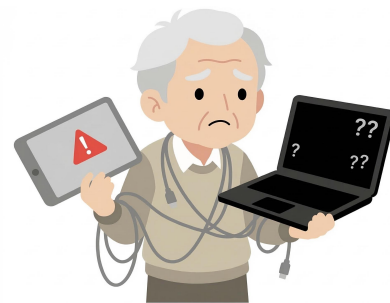
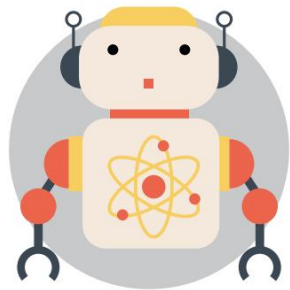


基于大模型智能体的老年人数字抵触形成 与演进机制研究

从数字摩擦到习得性无助的
社会计算



项目背景

数字摩擦对老年人信心的摧毁与习得性无助现象



01

数字摩擦的累积效应 —

交互挫败不断累积，导致老年人产生不可控感。

02

自身消极的归因偏差 —

倾向于内在、稳定、全局的解释风格

将可能的设计缺陷错误归结为“老了、笨、时代变了”。

03

负面预期的心理闭环 —

确信努力也无法改变未来结果

04

产生习得性无助 —

How older people account for their experiences with interactive technology

289

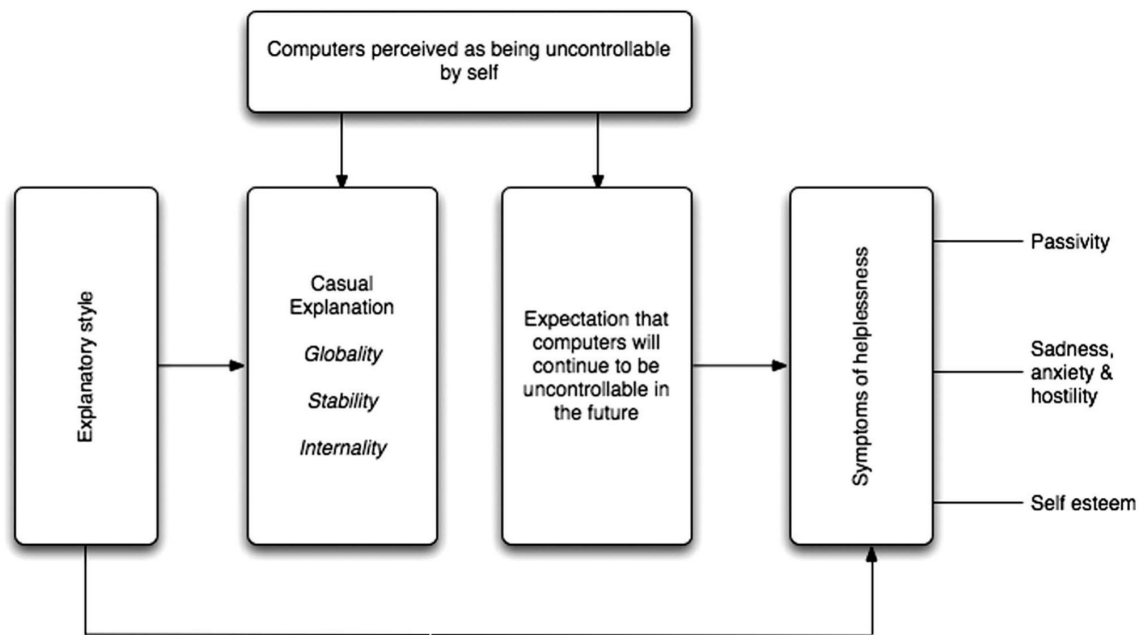


Figure 1. Explanatory style, causal explanations and learned helplessness in relation to learning to use computers.

Turner, Phil & Turner, Susan & Van de Walle, Guy. (2007). How older people account for their experiences with interactive technology. *Behaviour & Information Technology*. 26. 287–296. 10.1080/01449290601173499.



项目背景

传统研究方法的局限性

01 | 现实社会实验的困境

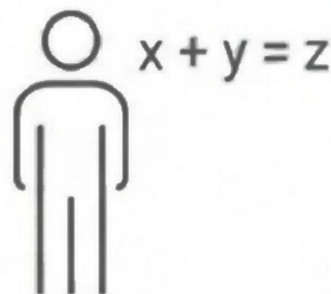


极高的成本与时间跨度



伦理与道德上的风险

02 | 传统Agent(ABM)的瓶颈 Underdeveloped Intelligence



基于规则、方程或机器学习模型驱动。

行为模式单一，缺乏人类认知的复杂性。

难以产生用于深层机制解释的定性数据。

项目背景

社会模拟计算的优势与可行性



较低的成本与时间跨度



不存在伦理与道德上的风险

01 | 模拟社会实验的优势

可展现出类似人类的非理性、情感决策。

02 | LLM-Based Agent的优势



可以提供更多的数据

Agent具备可被追问的心智，用于机制解释。

已有社会模拟平台的比较

Table 1: Comparison of LLM-driven agents and social simulators.

Model	Minds			Mobility		Economics		Social		Others		M-B		Scale	Env.
	RP.	AM.	SI.	Relo.	Traj.	Res.	Exc.	Int.	Rel.	Con.	Free	Infl.	Dri.	#	
D2A [101]	✓	✓	✓	✓						✓	✓			<100	Text
Ecoagent [64]	✓	✓				✓	✓					✓		100-1k	Rules
OASIS [109]	✓	✓						✓	✓			✓		>10k	Rules
GA1000 [78]	✓	✓	✓											1k-10k	×
MATRIX [75]	✓	✓						✓						<100	Text
Sociodojo [24]	✓					✓								<100	Data
GA [77]	✓	✓		✓	✓			✓	✓	✓	✓	✓	✓	<100	Rules
GenSim [94]	✓	✓								✓				>10k	Data
Project Sid [6]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	1k-10k	Rules
AgentScope [40]	✓	✓						✓						N/A	N/A
HiSim [72]	✓	✓						✓	✓					100-1k	Rules
S3	✓	✓						✓	✓					1k-10k	Rules
Agent4Rec [110]	✓	✓								✓				1k-10k	Rules
RecAgent [100]	✓	✓						✓	✓	✓				100-1k	Rules
Sotopia [116]	✓							✓						<100	×
Casevo [52]	✓	✓						✓	✓					100-1k	Rules
Ours	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	>10k	Society

Piao, Jinghua & Yan, Yuwei & Zhang, Jun & Li, Nian & Yan, Junbo & Lan, Xiaochong & Lu, Zhihong & Zheng, Zhiheng & Wang, Jing Yi & Zhou, Di & Gao, Chen & Xu, Fengli & Zhang, Fang & Rong, Ke & Su, Jun & Li, Yong. (2025). AgentSociety: Large-Scale Simulation of LLM-Driven Generative Agents Advances Understanding of Human Behaviors and Society. arXiv. 2502.08691. <https://arxiv.org/abs/2502.08691>.

AgentSociety的优势

Silicon Participants

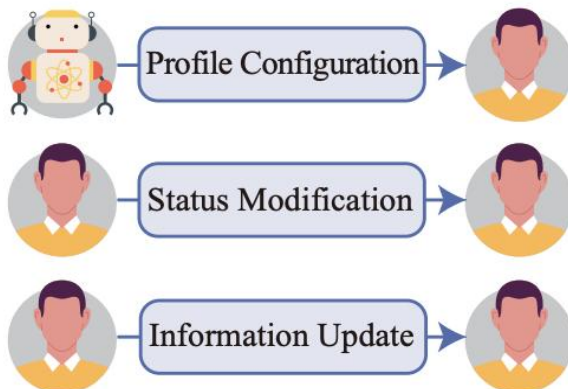


Profiles & Status

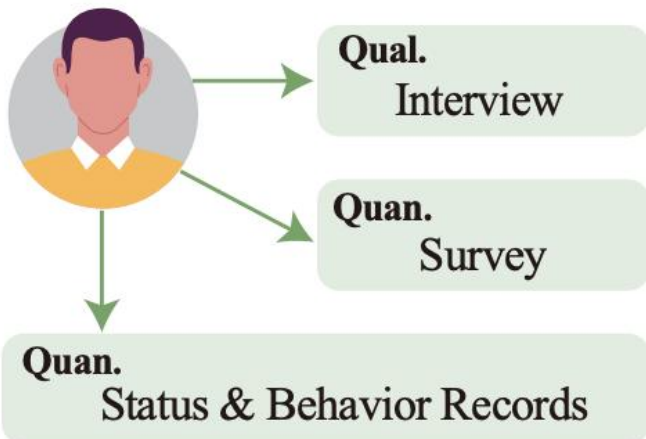
Memory & Minds

Social Behaviors

Interventions



Data Collection



01

Agent层面

LLM赋予了agent响应能力,可测量性,可追问性

02

实验流程上

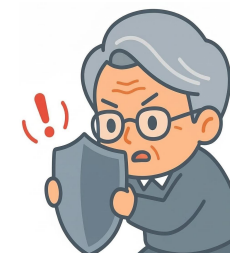
提供了一套完整的实验框架, 包含
Silicon Participants;
干预机制; 完整的数据收集工具链

03

基于社会科学理论的建模

保证了agent可解释性和可控性

项目设置Agent异质性建模

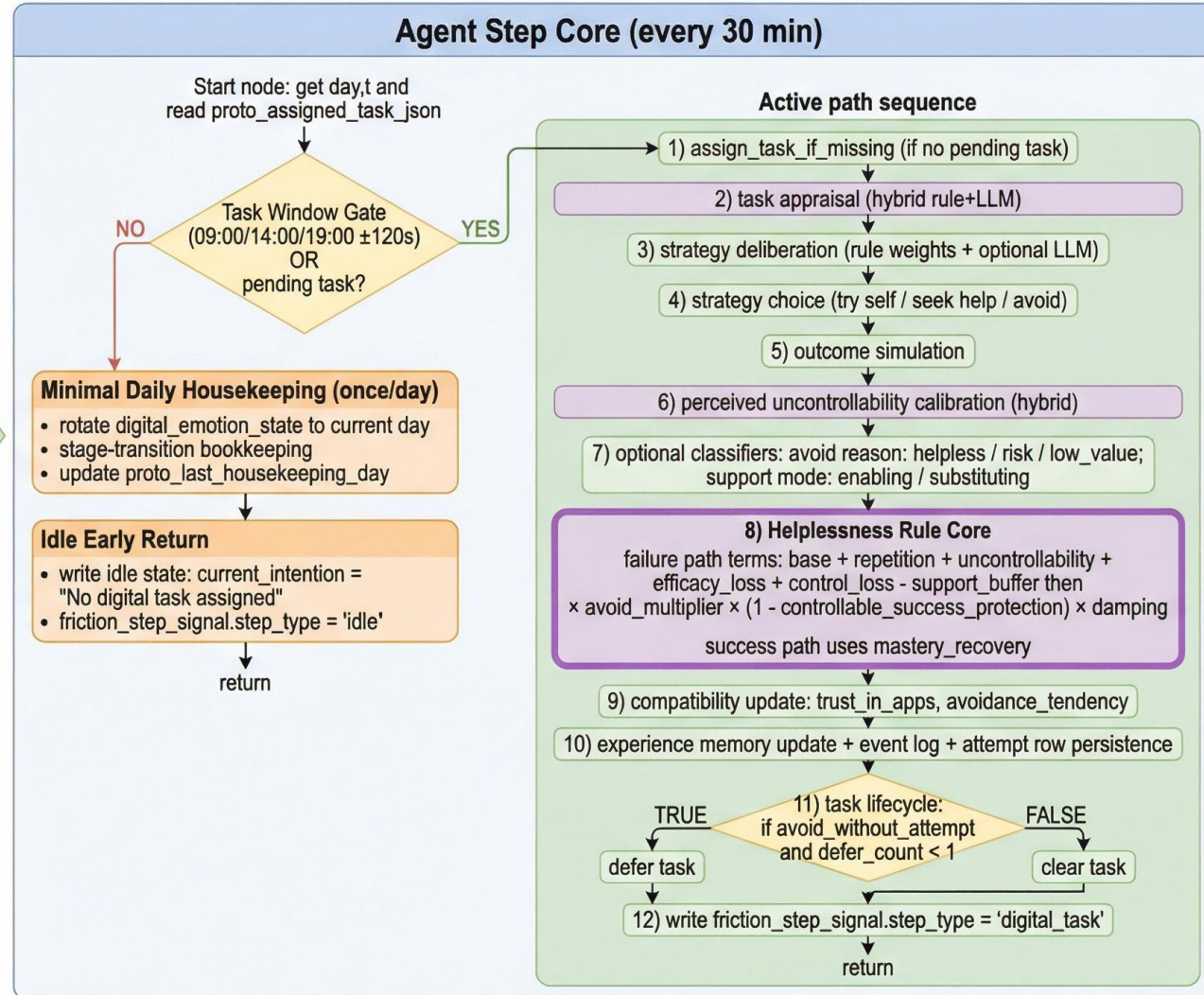
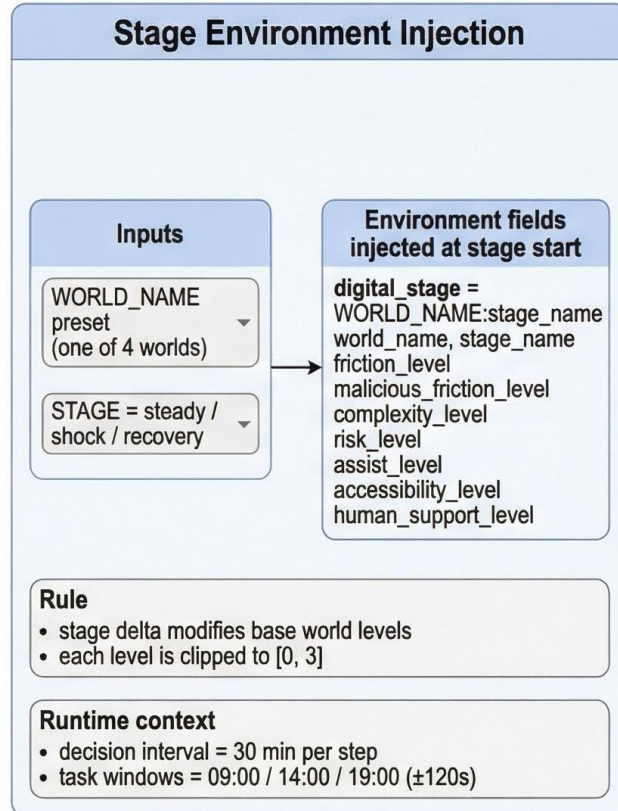


项目设置Agent异质性建模

```
{
  "id": 1,
  "name": "李春梅",
  "gender": "女",
  "age": 67,
  "education": "小学",
  "occupation": "退休",
  "persona": "谨慎务实者",
  "digital_experience": 0.2,
  "vision_limit": 0.6,
  "past_fraud_experience": 0.4,
  "background_story": "李春梅, 67岁, 退休后主要在社区活动, 平时喜欢去公园散步、参加社区合唱和健康讲座。她有一部用了多年的旧款手机, 日常只会接打电话和发微信, 偶尔刷短视频。视力下降让她对小字号和复杂页面特别不安。遇到验证码、弹窗或多步确认时容易焦虑, 常需要家人远程指导。她重视安全与省钱, 听到“优惠”“领奖”会本能警惕。"
},
{
  "id": 2,
  "name": "周德荣",
  "gender": "男",
  "age": 71,
  "education": "初中",
  "occupation": "退休",
  "persona": "社区纽带",
  "digital_experience": 0.4,
  "vision_limit": 0.3,
  "past_fraud_experience": 0.1,
  "background_story": "周德荣, 71岁, 长期担任楼栋志愿者, 熟悉社区事务, 邻里有事常找他帮忙协调。他会用手机支付、网约车和挂号, 但对频繁弹窗、默认勾选和验证码输入非常不耐烦。过去帮邻居处理过几次误扣费问题, 因此对“看不清的按钮”和“套路流程”很反感。遇到系统更新或页面改版时, 他往往选择求助子女或直接放弃。"
},
{
  "id": 3,
  "name": "王秀兰",
  "gender": "女",
  "age": 74,
  "education": "小学",
  "occupation": "退休",
  "persona": "保守谨慎者",
  "digital_experience": 0.1,
  "vision_limit": 0.7,
  "past_fraud_experience": 0.6,
  "background_story": "王秀兰, 74岁, 独居, 儿女在外地工作。几年前接到过“冒充客服退款”的电话, 虽然未被骗但受惊不小, 从此对线上支付极度敏感。她能用微信聊天和看新闻, 但涉及转账、验证码或银行卡绑定就会紧张, 常请求家人协助或干脆不用。视力不好使她对小字提示与复杂流程更不信任, 宁愿去线下窗口办理。"
```



DEMO流程



Agent可能的演进过程

阶段一：初期受挫

>_ Action: Attempt_Back → Trigger_Ad

Helplessness: 0.1  强

“这个广告图标和返回键太像了，下次我要更仔细一点。”

阶段二：认知震荡

>_ Action: Retry_Pwd_x3 → Failed

Helplessness: 0.5  动摇

“输入三遍都不对……是我记错了，还是我真的看不清这些字？”

阶段三：彻底无助

>_ Action: Encounter_Popup → Exit

Helplessness: 0.9  回避

“又来了。我怎么点都是错的，我这种老头子根本不适合用智能手机。不添麻烦了。”

DEMO结果

Table 1: Paired effects vs baseline in vertical layout (Stage 6, 10-day, 3-seed

Contrast vs baseline	Metric	Mean Δ	95% CI
High Friction + Low Assist	Attempt rate	-0.349	[-0.390, -0.301]
	Neg. share	+0.449	[+0.418, +0.472]
	Helplessness	+18.614	[+16.829, +20.143]
	Trust	-19.783	[-20.425, -19.400]
	Avoidance	+7.167	[+4.850, +8.350]
High Friction + High Assist	Attempt rate	-0.116	[-0.144, -0.078]
	Neg. share	+0.195	[+0.171, +0.219]
	Helplessness	+14.645	[+9.296, +18.175]
	Trust	-22.017	[-29.950, -14.575]
	Avoidance	+12.033	[+4.325, +17.650]
Low Friction + High Assist	Attempt rate	+0.109	[+0.093, +0.133]
	Neg. share	-0.170	[-0.181, -0.155]
	Helplessness	-27.158	[-30.085, -24.262]
	Trust	+40.083	[+34.625, +44.000]
	Avoidance	-35.408	[-41.475, -30.375]

A

小幅改善整体趋稳

B

高摩擦会显著拉低表现

C

高辅助难抵高摩擦损害

方向与理论预期一致。

项目预期成果

成熟的老年人数字抵触演化模型

可量化的“数字无助”指标体系

填补长期演化研究空白

不同画像下的最优干预策略



请老师批评指正